Mathematical Foundations of Football Quantitative Analytics and Sports Betting: The FootyQuant v2 Scientific Framework

Deliverable A: Executive Research Summary

Quantitative football modeling and sports betting market evaluation require a rigorous, multidisciplinary synthesis of probability theory, Bayesian statistics, stochastic processes, market microstructure, and mathematical decision theory. Association football presents distinct analytical challenges: goal scoring is a low-frequency, discrete stochastic process subject to temporal non-stationarity, tactical interactions, and non-linear dependencies between competing teams. Furthermore, betting markets function as complex financial exchanges characterized by asymmetric information, bookmaker margins, adverse selection, and behavioral biases such as the favourite-longshot effect.

This research report establishes the mathematical, statistical, and economic foundation for the FootyQuant v2 decision-support system. Rather than treating prediction and capital allocation as separate problems, this investigation demonstrates that probability estimation, forecast calibration, market margin extraction, expected utility evaluation, and portfolio risk management form an interdependent chain. An error in probability calibration directly distorts expected value calculations, while unquantified parameter variance in expected value estimation leads to severe capital over-allocation and heightened risk of ruin under standard Kelly criterion staking.

The primary conclusions of this investigation are:

1. **Goal Distribution Non-Independence and Low-Score Correction**: The independent Poisson model introduced by Maher (1982) serves as an essential baseline but systematically misestimates low-scoring draw and win probabilities. The Dixon-Coles model (1997) and bivariate Poisson formulations (Karlis & Ntzoufras, 2003) resolve this failure mode by introducing structural low-score dependency parameters (\rho or \lambda_3) and dynamic time-decay weighting functions.
2. **Forecast Calibration Over Pure Discrimination**: Evaluating betting models using classification metrics such as ROC-AUC or accuracy is mathematically flawed. Optimal decision-making under uncertainty depends entirely on probability calibration. Models must be evaluated using strictly proper scoring rules, primarily Log Loss and Brier Score decomposition, with post-hoc probability calibration achieved through Beta calibration.
3. **Market Microstructure and Margin De-vigging**: Bookmaker decimal odds contain systematic margins and risk-aversion adjustments. Proportional and additive de-vigging methods fail to account for adverse selection. Shin's method (1991, 1993) provides a mathematically defensible model that recovers fair, un-vidded probabilities by explicitly accounting for the proportion of informed traders in the market.
4. **Parameter Uncertainty and Fractional Staking**: The standard Kelly criterion maximizes long-run logarithmic wealth growth under the strict assumption that true probability is known with certainty. When applied to estimated probabilities subject to epistemic

variance, full Kelly staking leads to catastrophic drawdowns and deterministic ruin. Capital allocation must incorporate model uncertainty via Bayesian posterior predictive distributions and fractional Kelly adjustments ($c \in [0.10, 0.25]$).

5. **Portfolio Risk and Correlated Wagers**: Evaluating multiple simultaneous wagers as independent wagers overestimates diversification and underestimates aggregate exposure. A complete portfolio optimization framework must account for covariance across markets within the same match and common-factor risk across fixtures in the same league.

Deliverable B: Mathematical Foundations Handbook

1. Probability Theory and Stochastic Foundations

Kolmogorov Axioms and Probability Spaces

A formal probability model is defined on a probability space $(\Omega, \mathcal{F}, P)$, where Ω represents the sample space containing all possible match scorelines or event sequences, $\mathcal{F}$ is a σ-algebra of events (subsets of Ω), and $P: \mathcal{F} \to [0, 1]$ is a probability measure satisfying Kolmogorov's axioms:

1. **Non-negativity**: $P(A) \ge 0, \quad \forall A \in \mathcal{F}$
2. **Unitarity**: $P(\Omega) = 1$
3. **Countable Additivity**: For any sequence of mutually disjoint sets $A_1, A_2, \dots \in \mathcal{F}$: $P\left(\bigcup_{i=1}^\infty A_i \right) = \sum_{i=1}^\infty P(A_i)$

In association football, Ω can represent the countable set of pairs $(x, y) \in \mathbb{N}_0 \times \mathbb{N}_0$, where x denotes home goals and y denotes away goals.

Conditional Probability and Bayes' Theorem

Conditional probability measures the likelihood of event A given the occurrence of event B (where $P(B) > 0$):

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Two events A and B are independent if and only if $P(A \cap B) = P(A)P(B)$. In football analytics, goal-scoring counts between competing teams are rarely independent due to shared match state, tempo, and tactical interactions.

Bayes' theorem updates the probability of a hypothesis (parameter vector $\boldsymbol{\theta}$) given observed match data D:

$$P(\boldsymbol{\theta} | D) = \frac{P(D | \boldsymbol{\theta}) P(\boldsymbol{\theta})}{P(D)} = \frac{P(D | \boldsymbol{\theta}) P(\boldsymbol{\theta})}{\int_\Omega P(D | \boldsymbol{\theta}') P(\boldsymbol{\theta}') d\boldsymbol{\theta}'}$$

Where $P(\boldsymbol{\theta})$ is the prior distribution reflecting historical parameter estimates, $P(D | \boldsymbol{\theta})$ is the likelihood function of the match data given parameters, $P(\boldsymbol{\theta} | D)$ is the posterior distribution, and $P(D)$ is the marginal likelihood (evidence).

Statistical Inference Paradigms

Frequentist inference assumes $\boldsymbol{\theta}$ is a fixed, unknown vector estimated via

Maximum Likelihood Estimation (MLE):
$\hat{\boldsymbol{\theta}}_{\text{MLE}} = \arg\max_{\boldsymbol{\theta}} \sum_{i=1}^N \ln f(x_i | \boldsymbol{\theta})$
Bayesian inference treats $\boldsymbol{\theta}$ as a random variable, propagating parameter uncertainty directly into out-of-sample probabilistic forecasts via the posterior predictive distribution:
$P(y^* | D) = \int_{\Theta} P(y^* | \boldsymbol{\theta}) P(\boldsymbol{\theta} | D) d\boldsymbol{\theta}$
Where y^* represents the unobserved future match scoreline.

Core Discrete and Continuous Probability Distributions

Distribution	Probability Mass / Density Function	Expectation E[X]	Variance $\text{Var}(X)$	Football Application
Bernoulli	$P(X=k) = p^k (1-p)^{1-k}, \quad k \in \{0, 1\}$	p	$p(1-p)$	Match outcome indicator (e.g., Home Win vs Non-Home Win).
Binomial	$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$	$n p$	$n p (1-p)$	Penalty shootout conversion success in n attempts.
Poisson	$P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k \in \mathbb{N}_0$	λ	λ	Baseline goal count per match for rate λ.
Negative Binomial	$P(X=k) = \frac{\Gamma(k+r)}{k! \Gamma(r)} (1-p)^r p^k$	$\frac{r p}{1-p} = \mu$	$\mu + \frac{\mu^2}{r}$	Overdispersed goal counts across multiple leagues or fixtures.
Normal	$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$	μ	σ^2	Continuous latent rating differences in Elo or Glicko models.
Beta	$f(p) = \frac{1}{B(\alpha, \beta)} p^{\alpha-1} (1-p)^{\beta-1}$	$\frac{\alpha}{\alpha + \beta}$	$\frac{\alpha \beta}{(\alpha+\beta)^2 (\alpha+\beta+1)}$	Prior and posterior distribution over win/draw/loss probabilities.
Gamma	$f(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta \lambda}$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$	Conjugate prior distribution for Poisson goal-scoring intensity rates.

Joint Distributions and Dependence Modelling via Copulas

When modeling bivariate goal outcomes (X, Y), independence assumptions often fail. Sklar's Theorem states that any multivariate joint cumulative distribution function (CDF) F(x, y) can be expressed in terms of its marginal CDFs F_X(x) and F_Y(y) using a copula C:
F(x, y) = C(F_X(x), F_Y(y))
For discrete distributions such as goal counts, copulas allow the decoupling of individual team goal-scoring distributions from their dependence structure, enabling parameterizations that capture tail dependence during high-scoring or low-scoring match environments.

Parameter Estimation and Regularization

Maximum likelihood estimates for high-dimensional models (such as team-level attack and defense parameters across multiple divisions) can suffer from overfitting in small samples. Regularized likelihood estimation introduces penalty terms:
\hat{\boldsymbol{\theta}}_{\text{Reg}} = \arg\max_{\boldsymbol{\theta}} \left\{ \sum_{i=1}^N \ln f(x_i | \boldsymbol{\theta}) - \lambda_1 \|\boldsymbol{\theta}\|_1 - \lambda_2 \|\boldsymbol{\theta}\|_2^2 \right\}
Where \lambda_1 enforces sparsity (Lasso/L1 regularization) and \lambda_2 shrinks parameters toward zero (Ridge/L2 regularization), shrinking noisy early-season team ability estimates toward the league baseline.

Hypothesis Testing, Confidence Intervals, and Multiple Testing

When evaluating model parameter significance or betting strategy excess returns, frequentist hypothesis testing calculates a p-value under the null hypothesis H_0. In high-dimensional model selection or strategy search across hundreds of markets, multiple hypothesis testing dramatically inflates False Discovery Rates (FDR). The Benjamini-Hochberg procedure controls the FDR at level q by ranking m sorted p-values p_{(1)} \le p_{(2)} \le \dots \le p_{(m)} and identifying the largest k such that:

p_{(k)} \le \frac{k}{m} q

Statistical Biases and Artifacts

- **Selection Bias**: Conditioning model training only on specific leagues or historical matches where bookmaker data was available.
- **Survivorship Bias**: Testing strategies on teams currently in the top flight, ignoring historically demoted teams.
- **Simpson's Paradox**: A statistical trend observed within individual leagues (e.g., higher shot conversion rates for home teams) reversing when data is aggregated across heterogeneous leagues with varying baseline scoring rates.
- **Regression to the Mean**: Extreme early-season performances (e.g., a team converting 30\% of expected goals) naturally reverting toward the population mean \mu. The expected performance \hat{x}_{t+1} given sample performance x_t is modeled as: \hat{x}_{t+1} = \mu + r (x_t - \mu) Where r \in [0, 1] represents the reliability coefficient.

Information Theory and Proper Scoring Rules

Information theory provides the mathematical metric for evaluating probabilistic predictions. The Shannon Entropy H(X) of a discrete random variable measures its intrinsic uncertainty:
$$H(X) = -\sum_{i=1}^K P(x_i) \log_2 P(x_i)$$
A scoring rule S(P, y) assigns a numerical penalty to a predicted probability distribution P given observed outcome y. A scoring rule is **strictly proper** if its expected loss is uniquely minimized when the reported probability distribution P matches the true data-generating distribution Q:
$$\mathbb{E}_{Y \sim Q}[S(Q, Y)] < \mathbb{E}_{Y \sim Q}[S(P, Y)], \quad \forall P \neq Q$$
The two foundational proper scoring rules in sports analytics are Log Loss (Cross-Entropy Loss) and Brier Score:
$$\text{Log Loss: } L(P, y) = -\sum_{k=1}^K y_k \ln P_k \text{Brier Score: } BS(P, y) = \frac{1}{K} \sum_{k=1}^K (P_k - y_k)^2$$
Where $y_k \in \{0, 1\}$ is a binary indicator for whether outcome k occurred, and $\sum y_k = 1$.

2. Football Goal and Match Outcome Modelling

Baseline Independent Poisson Model

The univariate independent Poisson model introduced by Maher (1982) models the goals scored by Home Team i (X_{ij}) and Away Team j (Y_{ij}) in a match as independent Poisson random variables:
$$X_{ij} \sim \text{Poisson}(\lambda_{ij}), \quad Y_{ij} \sim \text{Po[span_1](start_span)[span_1](end_span)isson}(\mu_{ij})$$
The expected goal intensities λ_{ij} and μ_{ij} are parameterized using a log-linear model:
$$\ln \lambda_{ij} = \mu + \gamma + \alpha_i + \beta_j \ln \mu_{ij[span_2](start_span)[span_2](end_span)} = \mu + \alpha_j + \beta_i$$
Where μ is the baseline overall goal-scoring log-rate, γ is the home field advantage parameter, α_i, α_j are the attacking strengths of the home and away teams, and β_i, β_j are the defensive weakness parameters of the home and away teams.
To make parameters identifiable, sum-to-zero constraints are imposed:
$$\sum_{k=1}^N \alpha_k = 0, \quad \sum_{k=1}^N \beta_k = 0$$
The joint probability of scoreline (x, y) under independence is:
$$P(X_{ij} = x, Y_{ij} = y) = \left(\frac{\lambda_{ij}^x e^{-\lambda_{ij}}}{x!} [span_4](start_span)[span_4](end_span)\right) \left(\frac{\mu_{ij}^y e^{-\mu_{ij}}}{y!} \right)$$
Unadjusted Poisson models assume equality of conditional mean and variance ($\mathbb{E}[X] = \text{Var}(X) = \lambda$). In empirical match data, pooled goal distributions exhibit overdispersion ($\text{Var}(G) > \mathbb{E}[G]$) due to variance in goal expectation parameters across fixtures. Furthermore, independent Poisson models systematically underestimate low-scoring draws (0-0, 1-1) and low-scoring home/away wins (1-0, 0-1).

Dixon-Coles Low-Score Dependency Correction

To correct low-score bias without altering the marginal distributions for higher goal counts, Dixon and Coles (1997) introduced an adjusted joint probability mass function incorporating a dependency parameter ρ:
$$P(X_{ij} = x, Y_{ij} = y) = \tau_{\rho, \lambda_{ij}, \mu_{ij}}(x, y) \frac{\lambda_{ij}^x e^{-\lambda_{ij}}}{x!} \frac{\mu_{ij}^y e^{-\mu_{ij}}}{y!}$$
Where the adjustment factor $\tau_{\rho, \lambda, \mu}(x, y)$ modifies only four specific

low-scoring cells $(x, y) \in \{0, 1\} \times \{0, 1\}$:
$$\tau_{\rho, \lambda, \mu}(x, y) = \begin{cases} 1 - \lambda \mu \rho & \text{for } x=0, y=0 \\ 1 + \mu \rho & \text{for } x=1, y=0 \\ 1 + \lambda \rho & \text{for } x=0, y=1 \\ 1 - \rho & \text{for } x=1, y=1 \\ 1 & \text{for } x+y > 2 \end{cases}$$
When $\rho > 0$, the model increases the probability of 1-0 and 0-1 scorelines while reducing 0-0 and 1-1 scorelines relative to independent Poisson predictions. Conversely, a negative ρ shifts probability toward 0-0 and 1-1 scorelines.
Dixon and Coles also integrated temporal decay into parameter estimation, weighting historical match k played at time t_k relative to current time t using an exponential weighting function:
$$\phi(t - t_k) = e^{-\xi (t - t_k)}$$
Where $\xi > 0$ is a temporal decay parameter optimized via out-of-sample log-loss minimization.

Bivariate Poisson Goal Model

Karlis and Ntzoufras (2003) formulated a structural bivariate Poisson model that introduces positive covariance between goal counts directly into the stochastic process. Let W_1, W_2, W_3 be independent Poisson random variables:
$$W_1 \sim \text{Poisson}(\lambda_1 - \lambda_3), \quad W_2 \sim \text{Poisson}(\lambda_2 - \lambda_3), \quad W_3 \sim \text{Poisson}(\lambda_3)$$
Defining home goals $X = W_1 + W_3$ and away goals $Y = W_2 + W_3$, the joint PMF is:
$$P(X = x, Y = y) = e^{-(\lambda_1 + \lambda_2 - \lambda_3)} \frac{(\lambda_1 - \lambda_3)^x}{x[span_101](start_span)[span_101](end_span)!} \frac{(\lambda_2 - \lambda_3)^y}{y!} \sum_{k=0}^{\min(x, y)} \binom{x}{k} \binom{y}{k} k! \left(\frac{\lambda_3}{(\lambda_1 - \lambda_3)(\lambda_2 - \lambda_3)} \right)^k$$
Where $\text{Cov}(X, Y) = \lambda_3$. Marginal distributions remain Poisson with $\mathbb{E}[X] = \lambda_1$ and $\mathbb{E}[Y] = \lambda_2$.

Skellam Distribution and Goal Difference Modelling

For betting markets focused purely on goal differentials (e.g., Asian Handicap or 1X2 outcomes), modeling goal differences $Z = X - Y$ directly bypasses full scoreline joint distributions. Karlis and Ntzoufras (2009) demonstrated that if $X \sim \text{Poisson}(\lambda_1)$ and $Y \sim \text{Poisson}(\lambda_2)$ are independent or bivariate Poisson distributed, their difference Z follows a Skellam distribution:
$$P(Z = z \mid \lambda_1, \lambda_2) = e^{-(\lambda_1 + \lambda_2)} \left(\frac{\lambda_1}{\lambda_2} \right)^{z/2} I_{|z|}\left(2 \sqrt{\lambda_1 \lambda_2} \right), \quad z \in \mathbb{Z}$$
Where $I_r(x)$ is the modified Bessel function of the first kind of order r:
$$I_r(x) = \sum_{k=0}^\infty \frac{1}{k! \, \Gamma(r + k + 1)} \left(\frac{x}{2} \right)^{2k + r}$$
To account for inflated draw frequencies in competitive leagues, a Zero-Inflated Skellam Model (ZISM) adds a point mass parameter p at z=0:
$$P(Z = z \mid \lambda_1, \lambda_2, p) = p \, \delta_{z,0} + (1 - p) P_{\text{Skellam}}(Z = z \mid \lambda_1, \lambda_2)$$
Where $\delta_{z,0} = 1$ if z=0, and 0 otherwise.

Bayesian Hierarchical Parameter Estimation

To stabilize parameter estimates in leagues with few played matches or across

promotion/relegation transitions, parameters are assigned hierarchical prior distributions. Implementation in Bayesian environments such as footBayes (Egidi et al.) employs half-Cauchy or normal hyperpriors:
\alpha_i \sim \mathcal{N}(0, \sigma_\alpha^2), \quad \beta_i \sim \mathcal{N}(0, \sigma_\beta^2) \sigma_\alpha \sim \text{Half-Cauchy}(0, \tau_\alpha), \quad \sigma_\beta \sim \text{Half-Cauchy}(0, \tau_\beta)
This hierarchy shrinks noisy team parameters toward zero (the league average), mitigating parameter variance early in a season.

Dynamic Rating Systems and State-Space Updating

Static parameters fail to track evolution in team performance caused by tactical changes, injuries, or transfer windows. Dynamic rating systems track latent team strength over time:

- **Elo Rating System**: Updates home strength R_A and away strength R_B after match k: R_A' = R_A + K \cdot G \cdot (S_A - E_A) Where K is the update weight, G is a margin-of-victory multiplier (G = \sqrt{\vert{}\Delta \text{goals}\vert{}}), S_A \in \{1, 0.5, 0\} is the observed result, and E_A is the expected win probability: E_A = \frac{1}{1 + 10^{-(R_A - R_B + H) / 400}} Where H is the home advantage constant.
- **State-Space Dynamic Models**: Model team attack \alpha_{i, t} and defense \beta_{i, t} as random walks: \alpha_{i, t} = \alpha_{i, t-1} + \epsilon_{i, t}^\alpha, \quad \epsilon_{i, t}^\alpha \sim \mathcal{N}(0, \sigma_w^2) \beta_{i, t} = \beta_{i, t-1} + \epsilon_{i, t}^\beta, \quad \epsilon_{i, t}^\beta \sim \mathcal{N}(0, \sigma_w^2) Sequential Bayesian updating is conducted via Kalman Filtering or sequential Monte Carlo (Particle Filtering).

Advanced Spatial Metrics: Expected Goals (xG) and Expected Threat (xT)

Shot-based metrics resolve the low-sample variance of actual goals scored by measuring underlying chance creation:

- **Expected Goals (xG)**: Models the conditional probability that shot m results in a goal given feature vector \mathbf{x}_m (distance, angle, body part, pass type, defender proximity): \text{xG}_m = P(\text{Goal}_m = 1 | \mathbf{x}_m) = \frac{1}{1 + e^{-\boldsymbol{\beta}^T \mathbf{x}_m}}
- **Expected Threat (xT)**: Introduces a spatial grid model to value ball progression (passes and carries) prior to shots (Karun Singh, 2018). The pitch is discretized into an M \times N grid (typically 16[span_179](start_span)[span_179](end_span)[span_180](start_span)[span_180](end_span) \times 12). For each cell (x, y), the threat value q_{x, y} is defined via a Markov decision recursion solved using Bellman value iteration: q_{x, y} = \left(s_{x, y} \cdot g_{x, y} \right) + \left(m_{x, y} \sum_{x'=1}^M \sum_{y'=1}^N T_{(x, y) \to (x', y')} \, q_{x', y'} \right) Where s_{x, y} is the probability of taking a shot from cell (x, y), g_{x, y} is the average shot conversion probability (\text{xG}) from cell (x, y), m_{x[span_182](start_span)[span_182](end_span)[span_188](start_span)[span_188](end_span), y} = 1 - s_{x, y} is the probability of moving the ball (passing or carrying) from cell (x, y), and [span_183](start_span)[span_183](end_span)[span_189](start_span)[span_189](end_span)T_{(x, y) \to (x', y')} is the spatial transition matrix specifying the probability of moving from cell (x, y) to cell (x', y') given a move action.The net value added by a pass or carry starting at (x_{\text{start}}, y_{\text{start}}) and ending at (x_{\text{end}},

y_{\tex[span_196](start_span)[span_196](end_span)[span_201](start_span)[span_201](end_span)t{end}}) is: \Delta \text{xT} = q_{x_{\text{end}}, y_{\text{end}}} - q_{x_{\text{start}}, y_{\text{start}}}

Deriving Betting Market Probabilities from Joint Score Distributions

Given a discrete joint goal mass matrix P(X=x, Y=y) calculated up to a cutoff N_{\max} = 15 where \sum_{x=0}^{N_{\max}} \sum_{y=0}^{N_{\max}} P(X=x, Y=y) \approx 1, probabilities for all primary betting markets are computed via exact summation over scoreline subsets:

1. **1X2 (Match Outcome)**: P(\text{Home Win}) = \sum_{x > y} P(x, y), \quad P(\text{Draw}) = \sum_{x = y} P(x, y), \quad P(\text{Away Win}) = \sum_{x < y} P(x, y)
2. **Over / Under K.5 Goals**: P(\text{Over } K.5) = \sum_{x+y > K} P(x, y), \quad P(\text{Under } K.5) = \sum_{x+y \le K} P(x, y)
3. **Both Teams to Score (BTTS)**: P(\text{BTTS Yes}) = \sum_{x \ge 1} \sum_{y \ge 1} P(x, y), \quad P(\text{BTTS No}) = 1 - P(\text{BTTS Yes})
4. **Asian Handicap (H - a)**: For home team handicap +a (where a \in \mathbb{R}): P(\text{AH Win}) = \sum_{x + a > y} P(x, y), \quad P(\text{AH Push}) = \sum_{x + a = y} P(x, y), \quad P(\text{AH Loss}) = \sum_{x + a < y} P(x, y)

3. Probabilistic Forecasting, Evaluation, and Calibration

Calibration versus Discrimination

Model evaluation must distinguish between ranking capability and probability accuracy:

- **Discrimination**: The ability of a model to assign higher probabilities to events that occur than to events that do not. Measured by Receiver Operating Characteristic - Area Under Curve (ROC-AUC). ROC-AUC is rank-based and invariant to monotonic probability transformations; a model predicting probabilities 0.90 and 0.99 for events with true frequencies 0.51 and 0.52 achieves perfect ROC-AUC despite producing completely invalid probabilities for expected value calculations.
- **Calibration**: The degree to which predicted probabilities match empirical realization frequencies. A forecast is calibrated if: P(Y = 1 | \hat{P} = p) = p, \quad \forall p \in [0, 1]

Proper Scoring Rule Decompositions

The Brier Score for multi-class outcome K over N matches admits a formal mathematical decomposition into Reliability, Resolution, and Uncertainty:
BS = \frac{1}{N} \sum_{k=1}^K \sum_{i=1}^N (f_{ik} - y_{ik})^2 = \text{Reliability} - \text{Resolution} + \text{Uncertainty}
Where forecasts are grouped into M probability bins with sample sizes n_j:
\text{Reliability} = \frac{1}{N} \sum_{j=1}^M n_j (\bar{f}_j - \bar{y}_j)^2 \\ \text{Resolution} = \frac{1}{N} \sum_{j=1}^M n_j (\bar{y}_j - \bar{y})^2 \\ \text{Uncertainty} = \bar{y} (1 - \bar{y})
Reliability directly quantifies calibration error (lower is better), Resolution measures class-separation capability (higher is better), and Uncertainty quantifies intrinsic environment variance.

Parametric and Non-Parametric Calibration Methods

Raw model output probabilities \hat{p} often exhibit overconfidence or systematic shift and require post-processing calibration:

1. **Platt Scaling**: Fits a logistic regression model over raw logit scores f(x): P_{\text{cal}}(Y=1 | f(x)) = \frac{1}{1 + e^{A f(x) + B}}
2. **Isotonic Regression**: Non-parametric piecewise constant step function fitted via the Pairwise Adjacent Violators Algorithm (PAVA). Risk of overfitting on small samples.
3. **Beta Calibration**: Recommended parametric calibration method for continuous probabilistic classifiers (Kull et al., 2017). Fits a Beta distribution ratio over raw probabilities r[span_206](start_span)[span_206](end_span) \in (0, 1): \ln \left(\frac{P_{\text{cal}}}{1 - P_{\text{cal}}} \right) = a \ln(r) + b \ln(1 - r) + c

Epistemic versus Aleatoric Uncertainty Quantification

Total forecast uncertainty consists of two distinct components:

- **Aleatoric Uncertainty**: Irreducible randomness inherent to the football match process (e.g., deflection goals, referee decisions). Represented by the variance of the observational distribution P(Y \vert{} \bo[span_210](start_span)[span_210](end_span)ldsymbol{\theta}).
- **Epistemic Uncertainty**: Reducible uncertainty stemming from parameter variance and limited training data. Represented by the variance of parameter posterior distributions P(\boldsy[span_212](start_span)[span_212](end_span)mbol{\theta} \vert{} D).

Using Bayesian ensemble sampling with S parameter samples \boldsymbol{\theta}^{(s)} \sim P(\boldsymbol{\theta} | D), the total predictive probability is:
\bar{P}(Y=1) = \frac{1}{S} \sum_{s=1}^S P(Y=1 | \boldsymbol{\theta}^{(s)})
The epistemic variance around the estimated probability is:
\text{Var}_{\text{epistemic}}(\hat{p}) = \frac{1}{S-1} \sum_{s=1}^S \left(P(Y=1 | \boldsymbol{\theta}^{(s)}) - \bar{P}(Y=1) \right)^2
When \text{Var}_{\text{epistemic}}(\hat{p}) is large, positive expected value decisions must be penalized or rejected.

4. Market Mathematics and Margin Removal

Odds Structures and Implied Probabilities

Decimal odds O_k represent the total payout per unit stake if outcome k occurs. The raw implied probability \pi_k is:
\pi_k = \frac{1}{O_k}
For a complete set of K mutually exclusive and exhaustive market outcomes (e.g., K=3 for 1X2):
S = \sum_{k=1}^K \pi_k = 1 + M
Where S is the bookmaker overround and M > 0 represents the total bookmaker margin.

Margin Removal (De-vigging) Methodologies

Recovering fair probabilities p_k (where \sum p_k = 1) requires removing margin M:

1. **Proportional Normalization**: Assumes margin is distributed proportional to raw implied probabilities: p_k = \frac{\pi_k}{S} = \frac{\pi_k}{\sum_{j=1}^K \pi_j}
2. **Additive Normalization**: Assumes margin is distributed equally across all K outcomes: p_k = \pi_k - \frac{S - 1}{K} Additive normalization can produce negative probabilities for

high-odds outcomes (longshots).

3. **Power Method**: Solves for exponent k_{pow} such that: $\sum_{k=1}^K \pi_k^{1 / k_{\text{pow}}} = 1 \implies p_k = \pi_k^{1 / k_{\text{pow}}}$
4. **Shin's Method**: Structural model accounting for adverse selection and insider trading (Shin, 1991, 1993). Assumes a fraction $z \in (0, 1)$ of market participants possess insider knowledge, while 1-z are uninformed. The relationship between fair probability p_k, raw price π_k, and insider proportion z is: $\pi_k = (1 - z) \frac{p_k}{1 - z p_k} + z p_k$ Solving Shin's model yields the exact insider fraction parameter z: $z = \frac{\sum_{k=1}^K \sqrt{z_k^2 + 4(1-z_k)} - 2}{K - 2}$ Recovered fair probabilities p_k are invariant to favourite-longshot distortions.

The structural transition from raw bookmaker odds to un-vidded probabilities requires applying these de-vigging algorithms sequentially, as summarized below:

De-vigging Method	Operational Formula	Mathematical Mechanism	Treatment of Favourite-Longshot Bias
Proportional	$p_k = \pi_k / \sum \pi_j$	Scales probabilities uniformly by overround factor S.	Preserves existing market bias proportionately.
Additive	$p_k = \pi_k - (S-1)/K$	Subtracts equal probability fraction from each outcome.	Over-corrects longshots; risks negative probabilities.
Power	$p_k = \pi_k^{1/k_{\text{pow}}}$	Solves non-linear exponent k_{pow} for sum to unity.	Compress longshot probabilities non-linearly.
Shin's Model	$\pi_k = (1-z) \frac{p_k}{1-zp_k} + zp_k$	Solves insider trader fraction z under adverse selection.	Structurally eliminates bias driven by informed volume.

Market Efficiency and Closing Line Value (CLV)

The Efficient Market Hypothesis (EMH) applied to sports betting posits that closing odds O_{closing} incorporate all publicly available information. The Closing Line Value (CLV) measures the ratio of executed decimal odds O_{exec} to final closing fair odds $O_{\text{fair, closing}}$:

$\text{CLV} = \frac{O_{\text{exec}}}{O_{\text{fair,}}$
clo[span_63](start_span)[span_63](end_span)sing}} - 1

Empirical research establishes that beating the closing line ($\text{CLV} > 0$) is a statistically significant indicator of long-term expected edge, whereas short-term realized returns are dominated by sample variance.

Documented Market Inefficiencies

- **Favourite-Longshot Bias (FLB)**: Systematic market mispricing where longshots are priced at lower expected returns (overpriced odds) relative to favourites. Driven by bookmaker risk aversion against longshot insider accumulation and bettor prospect-theory preferences for high-skewness payouts.
- **Line Movement and Sharp vs Recreational Flow**: Early market odds reflect bookmaker initial pricing models. Price movement toward closing is driven by high-capital institutional

syndicates ("sharp money"), causing closing prices to converge to highly efficient probabilities.

5. Expected Value and Decision Theory

Expected Value Derivation

The Expected Value (EV) of a decimal odds wager O with estimated outcome probability p is:
EV = p (O - 1) - (1 - p) = p O - 1
The break-even probability p_{\text{be}} is:
p_{\text{be}} = \frac{1}{O}
A wager possesses a positive mathematical edge if and only if p > p_{\text{be}}, yielding EV > 0.

Utility Theory and Risk Aversion

Decision-making under capital constraints replaces direct EV maximization with Expected Utility maximization. Under Von Neumann-Morgenstern utility with Constant Relative Risk Aversion (CRRA):
U(W) = \begin{cases} \ln(W) & \text{for } \gamma = 1 \\ \frac{W^{1-\gamma}}{1-\gamma} & \text{for } \gamma \neq 1 \end{cases}
Where W is total wealth and \gamma \ge 0 is the coefficient of relative risk aversion. Logarithmic utility (\gamma = 1) corresponds directly to long-run wealth growth rate optimization.

Bayesian Decision Theory and Asymmetric Costs

In sports betting, decision costs are asymmetric: placing a losing wager results in immediate capital loss (-1 unit stake), whereas abstaining from a positive edge wager results in an opportunity cost. Formally, a decision policy \delta(\mathbf{x}) \in \{\text{Bet}, \text{No Bet}\} selects action a to minimize expected loss under parameter uncertainty:

\delta^*(\mathbf{x}) = \arg\min_{a \in \mathcal{A}} \mathbb{E}_{\boldsymbol{\theta} | \mathbf{x}} \left[L(a, \boldsymbol{\theta}) \right]

Robust Decision Making and Abstention Policies

When parameter epistemic variance \text{Var}(\hat{p}) is high, point estimates of EV are unreliable. A robust decision rule enforces a dual-threshold abstention policy:
\text{Action} = \begin{cases} \text{Bet Stake } S & \text{if } \mathbb{E}[\hat{p}] \cdot O - 1 > EV_{\min} \quad \text{AND} \quad P\left(\hat{p} > \frac{1}{O} \right) \ge 1 - \alpha \\ \text{No Bet} & \text{otherwise} \end{cases}
Where 1 - \alpha (typically 0.90 or 0.95) is the required lower bound probability that the true win probability exceeds break-even.

6. Capital Allocation, Staking, and Risk Management

Derivation of the Kelly Criterion

Consider a repeatable wagering game with decimal odds O (net payout b = O - 1) and win probability p. Let f \in [0, 1) represent the fraction of current bankroll W_k staked on trial k.

Wealth updates as:
$W_k = \begin{cases} W_{k-1}(1 + f b) & \text{with probability } p \\ W_{k-1}(1 - f) & \text{with probability } q = 1 - p \end{cases}$
After N independent trials with S wins and N-S losses, total wealth is:
$W_N = W_0 (1 + f b)^S (1 - f)^{N-S}$
The exponential growth rate per trial g(f) is:
$g(f) = \lim_{N \to \infty} \frac{1}{N} \ln \left(\frac{W_N}{W_0} \right) = \mathbb{E}\left[\ln \left(\frac{W_k}{W_{k-1}} \right) \right] = p \ln(1 + f b) + (1 - p) \ln(1 - f)$
To maximize the growth rate, set the first derivative $\frac{d g(f)}{d f} = 0$:
$\frac{d g(f)}{d f} = \frac{p b}{1 + f b} - \frac{1 - p}{1 - f} = 0$ $p b (1 - f) = (1 - p)(1 + f b)$ \\ $p b - p b f = 1 - p + f b - p f b$ \\ $f^* = \frac{p b - (1 - p)}{b} = \frac{p(b + 1) - 1}{b} = \frac{p O - 1}{O - 1} = \frac{EV}{O - 1}$
Thus, the optimal full Kelly fraction f^* is exactly the expected value divided by net odds.
The Kelly growth curve g(f) is highly concave and strictly asymmetrical around f^*. Beginning at f=0, expected growth g(f) increases monotonically with stake fraction up to the peak optimal fraction f^*, where maximum logarithmic wealth accumulation $g_{\ma[span_216](start_span)[span_216](end_span)x} = g(f^*)$ is achieved. Staking within the fractional Kelly region ($0 < f < f^*$) sacrifices marginal growth in exchange for quadratic reductions in variance. Once stake size exceeds f^*, further increases cause expected growth g(f) to decline precipitously. At the critical threshold $f_{\text{over}} = 2 f^*$, expected long-run growth drops to zero ($g(2f^*) = 0$). Any stake fraction exceeding $2 f^*$ places the system in the over-betting collapse zone ($g(f) < 0$), where bankroll depletion occurs asymptotically with probability 1 despite a positive expected value per wager.

Fractional Kelly and Estimation Parameter Risk

In practical football betting, true probability p is unobserved and estimated as $\hat{p} = p + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma_e^2)$. Staking full Kelly under estimated $\hat{p}$ causes severe over-staking whenever $\epsilon > 0$. Fractional Kelly sets actual stake fraction $f_{\text{act}} = c \cdot f^*$, where $c \in (0, 1)$.

- For c = 0.50 (Half-Kelly): Retains 75\% of maximum growth while reducing bankroll variance by 50\%.
- For c = 0.25 (Quarter-Kelly): Standard institutional allocation choice to absorb parameter uncertainty and model misspecification.

Risk of Ruin and Drawdown Distributions

For a fixed bankroll B and flat wager sizes with unit edge e and return variance σ^2, the continuous approximation for Risk of Ruin R(B) is:
$R(B) = e^{-\frac{2 e B}{\sigma^2}}$
Under fractional Kelly staking, absolute ruin ($W \to 0$) is asymptotically zero in continuous space, but probability of experiencing a specified drawdown threshold $D \in (0, 1)$ (e.g., 50\% bankroll loss) before reaching target wealth W_{target} is:
$P(\text{Drawdown } \ge D) = D^{\frac{2}{c} - 1}$
For Full Kelly (c=1), $P(\text{Drawdown } \ge 0.50) = 0.50^{1} = 50\%$. For Quarter-Kelly (c=0.25), $P(\text{Drawdown } \ge 0.50) = 0.50^{7} = 0.78\%$.

Analytical Staking Comparison: ₹500 Initial Capital Example

To evaluate the mathematical mechanics of capital allocation, consider a starting bankroll W_0 = \text{₹}500. Assume a sequence of independent wagers offering decimal odds O = 2.00 (b = 1.0), where the true win probability is p = 0.55.

- **Expected Value**: EV = 0.55(2.00) - 1 = +0.10 (10\% edge).
- **Full Kelly Fraction**: f^* = \frac{0.10}{1.0} = 0.10 (10\% of current wealth). First stake: \text{₹}50.00.
- **Half Kelly Fraction (c=0.5)**: f = 0.05 (5\% of current wealth). First stake: \text{₹}25.00.
- **Quarter Kelly Fraction (c=0.25)**: f = 0.025 (2.5\% of current wealth). First stake: \text{₹}12.50.
- **Fixed Flat Stake (2\%)**: First stake: \text{₹}10.00.

Now evaluate performance under a parameter estimation error scenario where the model misestimates true probability as \hat{p} = 0.62, while true probability is actually p = 0.48 (Negative Edge EV = -0.04):

- Model calculated f^* = \frac{0.62(2.0) - 1}{1.0} = 0.24 (24\% stake).
- **Full Kelly**: Stakes \text{₹}120.00 on Bet 1. Long-run expected wealth growth rate g(f) = 0.48 \ln(1 + 0.24) + 0.52 \ln(1 - 0.24) = 0.48(0.2151) + 0.52(-0.2744) = -0.0394. The bankroll decays exponentially toward ruin.
- **Quarter Kelly (c=0.25)**: Stakes 0.25 \times 0.24 = 0.06 (6\% stake). Expected growth rate g(f) = 0.48 \ln(1.06) + 0.52 \ln(0.94) = 0.48(0.0583) + 0.52(-0.0619) = -0.0042. Capital loss rate is reduced by an order of magnitude.

Allocation Strategy	Stake Fraction Formula	First Stake (W_0=\text{₹}500, \hat{p}=0.55)	Growth Rate g(f) (True p=0.55)	50\% Drawdown Probability	Impact of +0.07 Parameter Estimation Error (\hat{p}=0.62, p=0.48)
Full Kelly (c=1.0)	f = \frac{\hat{p}O - 1}{O - 1}	\text{₹}50.00 (10.0\%)	0.00503	50.0\%	Extreme Decay (g = -0.0394)
Half Kelly (c=0.5)	f = 0.5 \cdot \frac{\hat{p}O - 1}{O - 1}	\text{₹}25.00 (5.0\%)	0.00379	3.12\%	Moderate Decay (g = -0.0102)
Quarter Kelly (c=0.25)	f = 0.25 \cdot \frac{\hat{p}O - 1}{O - 1}	\text{₹}12.50 (2.5\%)	0.00216	0.78\%	Minor Decay (g = -0.0042)
Fixed Flat (2\%)	f = 0.02	\text{₹}10.00 (2.0\%)	0.00178	N/A (Fixed size)	Linear Loss (-0.04 \times \text{Stake})

Splitting capital into discrete "safe" and "risky" sub-accounts is mathematically sub-optimal. Under log-utility, portfolio capital allocation is a single continuous constrained optimization problem. Artificially segregating capital alters effective risk tolerance non-linearly without altering the underlying risk-return frontier.

7. Portfolio Theory and Correlated Football Bets

Multi-Asset Kelly Optimization

When placing M simultaneous wagers on different matches or markets, independent Kelly allocation over-allocates total capital. Let $\mathbf{r} = (r_1, r_2, \dots, r_M)^T$ be the random vector of net wager returns per unit stake, with expected return vector $\mathbf{e} = \mathbb{E}[\mathbf{r}]$ and return covariance matrix $\boldsymbol{\Sigma} = \mathbb{E}[(\mathbf{r} - \mathbf{e})(\mathbf{r} - \mathbf{e})^T]$.
The multi-asset optimal Kelly stake vector $\mathbf{f}^* = (f_1, f_2, \dots, f_M)^T$ maximizes expected portfolio log-growth:
$$\max_{\mathbf{f}} \mathbb{E}\left[\ln\left(1 + \mathbf{f}^T \mathbf{r} \right) \right]$$
Using a second-order Taylor expansion approximation around $\mathbf{f} = \mathbf{0}$:
$$\mathbb{E}\left[\ln\left(1 + \mathbf{f}^T \mathbf{r} \right) \right] \approx \mathbf{f}^T \mathbf{e} - \frac{1}{2} \mathbf{f}^T \boldsymbol{\Sigma} \mathbf{f}$$
Setting the gradient to zero yields the matrix Kelly equation:
$$\mathbf{f}^* = \boldsymbol{\Sigma}^{-1} \mathbf{e}$$
Subject to aggregate leverage constraints:

$$\sum_{i=1}^M f_i \le f_{\max}, \quad f_i \ge 0, \quad \forall i$$

Within-Match Market Covariance

Wagers placed on multiple markets within the same match (e.g., Home Win and Over 2.5 Goals) exhibit strong non-zero return correlation. Let A be Home Win (O_A) and B be Over 2.5 Goals (O_B). The joint outcome probability distribution $P(A \cap B)$, $P(A \cap \neg B)$, $P(\neg A \cap B)$, $P(\neg A \cap \neg B)$ determines return covariance:
$$\text{Cov}(r_A, r_B) = \mathbb{E}[r_A r_B] - \mathbb{E}[r_A]\mathbb{E}[r_B]$$
Where:
$$\mathbb{E}[r_A r_B] = P(A \cap B)(O_A - 1)(O_B - 1) - P(A \cap \neg B)(O_A - 1) - P(\neg A \cap B)(O_B - 1) + P(\neg A \cap \neg B)$$
If positive covariance is ignored, separate univariate Kelly calculations double-count shared underlying variance, dramatically over-exposing capital to single-match structural outcomes.

8. Dynamic Event Modelling, Advanced Methods, and Data Reliability

In-Play Markov Chains and Point Processes

Live in-play betting requires modeling score updates dynamically. The match state is defined as a continuous-time Markov process $S(t) = (X(t), Y(t)) \in \mathbb{N}_0 \times \mathbb{N}_0$ for time $t \in [0, 90]$. Goal scoring follows a non-homogeneous Poisson process with time-dependent hazard intensity rates $\lambda_H(t \mid S(t))$ and $\mu_A(t \mid S(t))$:
$$\lambda_H(t) = \lambda_0(t) \exp\left(\boldsymbol{\beta}^T \mathbf{x}(t) + \delta_H \cdot I_{\text{Red Card}}(t) + \eta \cdot (X(t) - Y(t)) \right)$$
Where $\lambda_0(t)$ captures the empirical end-of-half intensity spike, $I_{\text{Red Card}}(t)$ models numerical player asymmetry, and $X(t) - Y(t)$ models defensive score-effects (teams leading tend to lower offensive pace).

Data Quality, Metric Reliability, and Data Leakage

Statistical predictions depend on data integrity and feature availability timestamps:

- **xG Provider Variance**: Provider methodologies differ significantly. Models trained on Opta xG cannot be evaluated directly against StatsBomb or Wyscout xG without standardizing coordinate definitions and shot-context features.
- **Data Leakage (Look-Ahead Bias)**: Severe validation failure occurring when future information is accidentally included in historical model features (e.g., using official lineup features for a match played 3 hours before official team sheet publication).
- **Timestamp Integrity**: All model features must be locked to historical time t_{\text{execution}}, defined as the exact second a market order is evaluated.

Deliverable C: Annotated Research Library

The primary literature powering FootyQuant v2 is documented below:

1. Primary Methodological Papers

Paper 1: Maher (1982)

- **Exact Title**: Modelling Association Football Scores
- **Author(s)**: Michael J. Maher
- **Publication Year**: 1982
- **Journal / Publisher**: *Statistica Neerlandica*, Vol. 36, Issue 3, pp. 109–118
- **DOI / Link**: https://doi.org/10.1111/j.1467-9574.1982.tb00782.x
- **Source Type**: Peer-reviewed journal article
- **Mathematical Topic**: Poisson regression, team attack/defense parameterization
- **Main Contribution**: Formulated the first parametric bivariate model fitting team-specific attacking strengths (\alpha) and defensive weaknesses (\beta) via independent Poisson goal processes.
- **Core Equations**: \lamb[span_6](start_span)[span_6](end_span)da_{ij} = \alpha_i \beta_j \gamma,[span_7](start_span)[span_7](end_span) \quad \mu_{ij} = \alpha_j \b[span_8](start_span)[span_8](end_span)eta_i
- **Empirical Evidence**: Tested on 1970s English Football League data; demonstrated statistically significant improvement over naive historic pooled goal distributions.
- **Limitations**: Assumes independence between home and away scorelines; fails to model overdispersion or low-score draw clustering.
- **Relevance**: Foundational Tier 1 baseline scoreline generator.
- **Priority**: Essential.

Paper 2: Dixon and Coles (1997)

- **Exact Title**: Modelling Association Football Scores and Inefficiencies in the Football Betting Market
- **Author(s)**: Mark J. Dixon and Stuart G. Coles
- **Publication Year**: 1997
- **Journal / Publisher**: *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, Vol. 46, Issue 2, pp. 265–280
- **DOI / Link**: https://doi.org/[span_224](start_span)[span_224](end_span)[span_229](start_span)[span

_229](end_span)10.1111/1467-9876.00065
- **Source Type**: Peer-reviewed journal article
- **Mathematical Topic**: Bivariate score adjustment, temporal weighting, maximum likelihood estimation
- **Main Contribution**: Introduced low-score dependency adjustment factor \tau_\rho(x,y) to fix 0-0, 1-0, 0-1, 1-1 scoreline probability deficits, alongside exponential time-decay parameter estimation \phi(t).
- **Core Equations**: P(X=x, Y=y) = \tau_{\rho, \lambda, \mu}(x,y) \, \frac{\lambda^x e^{-\lambda}}{x!} \frac{\mu^y e^{-\mu}}{y!}
- **Empirical Evidence**: Out-of-sample testing on 1995–1996 English league matches demonstrated positive betting returns against contemporary bookmaker odds.
- **Limitations**: Score adjustment \tau is restricted to low goal counts (x, y \le 1); does not capture non-linear structural shift in live matches.
- **Relevance**: Foundational core candidate model.
- **Priority**: Essential.

Paper 3: Karlis and Ntzoufras (2003)

- **Exact Title**: Analysis of sports data using bivariate Poisson models
- **Author(s)**: Dimitris Karlis and Ioannis Ntzoufras
- **Publication Year**: 2003
- **Journal / Publisher**: *Journal of the Royal Statistical Society: Series D (The Statistician)*, Vol. 52, Issue 3, pp. 381–393
- **DOI / Link**: https://doi.org/[span_244](start_span)[span_244](end_span)[span_245](start_span)[span_245](end_span)10.1111/1467-9884.00366
- **Source Type**: Peer-reviewed journal article
- **Mathematical Topic**: Bivariate Poisson distributions, joint score modeling
- **Main Contribution**: Formulated full joint bivariate Poisson distribution using a shared latent covariance parameter \lambda_3, replacing informal low-score adjustments with structural probability theory.
- **Core Equations**: X = W_1 + W_3, \quad Y = W_2 + W_3, \quad W_3 \sim \text{Poisson}(\lambda_3)
- **Empirical Evidence**: Fitted via EM algorithm and MCMC to football data; verified superior fit over independent Poisson models across multiple leagues.
- **Limitations**: Computational complexity is higher; fixed \lambda_3 assumes uniform goal covariance across all team match-ups.
- **Relevance**: Core probability engine for joint goal distributions.
- **Priority**: High Priority.

Paper 4: Karlis and Ntzoufras (2009)

- **Exact Title**: Bayesian modelling of football outcomes: using the Skellam's distribution for the goal difference
- **Author(s)**: Dimitris Karlis and Ioannis Ntzoufras
- **Publication Year**: 2009
- **Journal / Publisher**: *IMA Journal of Management Mathematics*, Vol. 20, Issue 2, pp. 133–145

- **DOI / Link**: https://doi.org/10.1093/imaman/dpn026
- **Source Type**: Peer-reviewed journal article
- **Mathematical Topic**: Skellam distribution, Zero-Inflated Skellam Model (ZISM), Bayesian MCMC
- **Main Contribution**: Introduced goal-difference modeling via the Skellam distribution, avoiding full joint scoreline estimation when targeting margin/handicap markets, with zero-inflation to handle excess draws.
- **Core Equations**: P(Z=z) =[span_146](start_span)[span_146](end_span) e^{-(\lambda_1+\lambda_2)} \left(\frac{\lambda_1}{\lambd[span_147](start_span)[span_147](end_span)a_2} \right)^{z/2} I_{|z|}(2\sqrt{\lambda_1\lambd[span_148](start_span)[span_148](end_span)a_2})
- **Empirical Evidence**: Bayesian posterior checks via MCMC demonstrated high accuracy in capturing draw probabilities and goal margins.
- **Limitations**: Discards individual total goal information (e.g., cannot price Over/Under 2.5 goals directly).
- **Relevance**: Essential specialized model for Asian Handicap and 1X2 valuation.
- **Priority**: High Priority.

Paper 5: Egidi, Karlis, and Ntzoufras (2022/2026)

- **Exact Title**: Predictive Modelling for Football Analytics / footBayes Package
- **Author(s)**: Leonardo Egidi, Dimitris Karlis, Ioannis Ntzoufras
- **Publication Year**: 2022 (R Package) / 2026 (Book, CRC Press)
- **Journal / Publisher**: CRC Press / Journal of Open Source Software
- **DOI / Link**: https://doi.or[span_258](start_span)[span_258](end_span)g/10.1201/9781003186496
- **Source Type**: Academic Monograph & R Software Package
- **Mathematical Topic**: Bayesian hierarchical modeling, Hamiltonian Monte Carlo, Stan * **Main Contribution**: Developed unified Bayesian framework for double Poisson, bivariate Poisson, and Skellam models incorporating student-t goal difference priors and dynamic team parameters.
- **Core Equations**: Implementation of MCMC sampling over parameter hyperpriors \alpha_i \sim \mathcal{N}(0, \sigma^2).
- **Empirical Evidence**: Extensive cross-validation across European top leagues demonstrating superior out-of-sample Brier scores compared to maximum likelihood baselines.
- **Limitations**: MCMC sampling requires significant computational infrastructure for real-time live betting updates.
- **Relevance**: Core computational Bayesian reference architecture.
- **Priority**: High Priority.

Paper 6: Shin (1991, 1993)

- **Exact Title**: Measuring the Incidence of Insider Trading in a Market for Betting / Optimal Betting Odds against Insider Traders
- **Author(s)**: Hyun Song Shin
- **Publication Year**: 1991, 1993
- **Journal / Publisher**: *The Economic Journal*, Vol. 101, pp. 1179–1185 / Vol. 103, pp.

1141–1153
- **DOI / Link**: https://doi.org/10.2307/2234434
- **Source Type**: Peer-reviewed journal article
- **Mathematical Topic**: Game theory, market microstructure, margin de-vigging
- **Main Contribution**: Solved the equilibrium bookmaker pricing problem under asymmetric information, deriving an analytical formula to recover true outcome probabilities by estimating insider trader proportion z.
- **Core Equations**: \pi_k = (1-z) \frac{p_k}{1 - z p_k} + z p_k
- **Empirical Evidence**: Validated on horse racing and European football betting markets; proved superior to proportional normalization in removing favourite-longshot bias.
- **Limitations**: Assumes uniform insider distribution across all market selections.
- **Relevance**: Essential foundational algorithm for market probability extraction.
- **Priority**: Essential.

Paper 7: Kull, de Menezes, and Flach (2017)

- **Exact Title**: Beta calibration: a well-calibrated without-loss-of-generality family of calibration models
- **Author(s)**: Meelis Kull, Miquel Perello-Nieto, Peter Flach
- **Publication Year**: 2017
- **Journal / Publisher**: *Advances in Neural Information Processing Systems (NeurIPS 30)*
- **DOI / Link**: https://proceedings.neurips.cc/paper/2017/hash/5103c3584b063c431bd1268e9b5e7642-Abstract.html
- **Source Type**: Peer-reviewed conference proceedings
- **Mathematical Topic**: Probability calibration, Beta distribution
- **Main Contribution**: Proved that Platt scaling distorts probabilities when scores are not normally distributed, establishing Beta Calibration as a mathematically robust parametric framework for classifiers bounded on (0, 1).
- **Core Equations**: \ln \frac{P_{\text{cal}}}{1 - P_{\text{cal}}} = a \ln(r) + b \ln(1 - r) + c
- **Empirical Evidence**: Demonstrated superior log loss and expected calibration error (ECE) across multi-class benchmark datasets compared to isotonic regression.
- **Limitations**: Requires numerical optimization for parameter bounds.
- **Relevance**: Core calibration algorithm for probability post-processing.
- **Priority**: High Priority.

Paper 8: Goddard (2004)

- **Exact Title**: Modelling Football Match Outcomes and the Efficiency of Fixed-Odds Betting Markets * **Author(s)**: John Goddard
- **Publication Year**: 2004
- **Journal / Publisher**: *International Journal of Forecasting*, Vol. 23, Issue 1, pp. 51–66
- **DOI / Link**: https://[span_268](start_span)[span_268](end_span)doi.org/10.1002/for.877
- **Source Type**: Peer-reviewed journal article
- **Mathematical Topic**: Ordered probit regression, market efficiency testing
- **Main Contribution**: Introduced ordered probit match outcome models incorporating extensive team performance metrics, geographical distance, and cup commitments to test weak-form market efficiency.

- **Core Equations**: Formulation of latent match outcome variable Y_i^* = \mathbf{x}_i^T \boldsymbol{\beta} + \epsilon_i with ordered threshold cuts \gamma_1, \gamma_2.
- **Empirical Evidence**: Extensive testing on 10 years of English League matches showing model capability to generate positive expected returns against specific historical bookmaker prices.
- **Limitations**: Goal-level scoreline distributions are not generated directly.
- **Relevance**: Comparative benchmark for direct 1X2 classification models.
- **Priority**: Useful Extension.

Deliverable D: Comparative Methods Matrix

The table below summarizes the quantitative methodologies investigated for FootyQuant v2:

Mathematical Method	Purpose	Required Inputs	Core Assumptions	Output	Math Complexity	Data Requirements	Interpretability	Failure Modes	System Tier
Independent Poisson (Maher)	Baseline Goal Scoring Engine	Historic match scorelines, team IDs	Goal counts are independent Poisson variables	Complete Joint Score Matrix	Low	Match results only	High	Underestimates 0-0, 1-1 draws; ignores goal overdispersion.	**Tier 1**
Dixon-Coles Model	Low-Score Dependent Scoring	Historical match results, match dates	Low goal counts (x,y \le 1) have non-linear correlation	Adjusted Joint Score Matrix	Medium	Match results + exact dates	High	Adjustment \tau is static for higher goal counts (x+y > 2).	**Tier 2**
Bivariate Poisson	Structural Correlated Goal Model	Match results, covariate metrics	Competing team goal counts share latent covariance \lambda_3	Correlated Joint Score Matrix	Medium-High	Match results, team metrics	Medium-High	Fixed covariance \lambda_3 may misvalue extreme mismatch games.	**Tier 2**
Zero-Inflated Skellam (ZISM)	Direct Goal Margin / Handicap Engine	Match results, handicap lines	Goal differentials follow Skellam distributi	Probability distribution over goal	High	Historic goal margins	Medium	Does not output total goals for Over/Un	**Tier 3**

Mathematical Method	Purpose	Required Inputs	Core Assumptions	Output	Math Complexity	Data Requirements	Interpretability	Failure Modes	System Tier
			on with draw inflation	differences Z				der markets.	
Bayesian Hierarchical Model	Regularized Team Strength Dynamics	Match scorelines, xG, team hyperpriors	Team ability parameters drawn from shared league distribution	Posterior probability score matrices	High	Multi-season match & xG data	High	High MCMC computational cost for live evaluation.	**Tier 2**
Expected Threat (xT) Grid	Spatial Ball Progression Valuation	Event data (pass/carry start/end locations)	Pitch value follows spatial Markov decision process	Spatial threat value grid q_{x,y}, \Delta\text{xT}	Medium-High	Spatial event data (coordinate logs)	High	Ignores defender positions and off-ball movement.	**Tier 2**
Gradient Boosted Trees (XGBoost/LightGBM)	Direct Market Classification	Tabular features (xG, ratings, rest, injuries)	Non-linear feature interactions without structural goal process	Binary/Multi-class probabilities	Medium	Extensive tabular feature store	Low (Black box)	Overfits small samples; produces uncalibrated tail probabilities.	**Tier 3**
Shin's De-vigging Method	Fair Market Probability Extraction	Bookmaker decimal odds across outcomes	Margin reflects adverse selection against insider fraction z	Unbiased fair market probabilities p_k	Medium	Multi-bookmaker odds feeds	High	Assumes uniform insider distribution across selections.	**Tier 1**
Beta Calibration	Probability Post-Processing	Uncalibrated probabilities, realized outcomes	Calibration curve follows Beta distribution log-odds	Fully calibrated probabilities	Medium	Validation set predictions + outcomes	Medium	Requires sufficient validation sample size (N > 500).	**Tier 2**

Mathematical Method	Purpose	Required Inputs	Core Assumptions	Output	Math Complexity	Data Requirements	Interpretability	Failure Modes	System Tier
			ratio						
Multi-Asset Portfolio Kelly	Optimal Multi-Bet Capital Allocation	Model probabilities, market odds, return covariance	Portfolio returns follow log-utility expansion	Stake vector \mathbf{f}^* across active wagers	High	Model outputs, market odds feeds	Medium	Matrix inversion \boldsymbol{\Sigma}^{-1} unstable if covariance singular.	**Tier 2**
Reinforcement Learning (RL) Staking	Dynamic Bankroll Control	Simulated market states, historical odds	Policy optimization via Q-learning/PPO	Dynamic stake action a_t	Very High	Millions of simulated match trials	Very Low	Highly susceptible to overfitting noisy historical paths.	**Tier 4**

Deliverable E: Evidence Assessment

Evaluating literature claims requires separating verified mathematical guarantees from empirical observations and speculative assumptions across a strict evidence hierarchy, as detailed below:

Evidence Level	Mathematical Domain	Validated Literature Findings	Boundaries and Critical Caveats
1. Mathematical Guarantees	Probability & Information Theory	Kolmogorov axioms, Bayes' rule updating, proper scoring rule expectation minimization (Log Loss, Brier Score), and Kelly growth optimization.	Guarantees hold strictly under assumed parameters; parameter estimation variance introduces risk.
2. Strong Empirical Support	Goal Distribution & Market Dynamics	Structural low-score draw inflation in football goal distributions, Closing Line Value (CLV) benchmark efficiency, and Favourite-Longshot Bias.	Empirical regularities persist across major leagues but exhibit structural shifts during rule changes.
3. Context-Dependent / Mixed Evidence	Machine Learning vs Structural Models	Performance comparison between tabular Gradient Boosted Trees (XGBoost) and	ML excels at capturing non-linear interactions but produces uncalibrated tail probabilities without

Evidence Level	Mathematical Domain	Validated Literature Findings	Boundaries and Critical Caveats
		structural statistical scoreline models (Dixon-Coles / Bivariate Poisson).	Beta calibration.
4. Unverified Claims / Speculation	Commercial Betting & Arbitrage Systems	Claims of guaranteed positive expected returns, backtested strategy persistence, and full Kelly staking survival in real markets.	Backtested profitability in small samples (N < 1000) is dominated by sample variance and overfit.

1. Mathematical Guarantees

- **Proper Scoring Rules**: Mathematically proven that minimizing expected Log Loss or Brier Score guarantees that predicted probabilities converge to true data-generating probabilities Q, provided the hypothesis space contains Q.
- **Kelly Growth Rate**: Mathematically proven that f^* = \frac{EV}{O - 1} yields the maximal asymptotic wealth growth rate \lim_{T \to \infty} \frac{1}{T} \ln W_T for a sequence of wagers with known, fixed probability p.

2. Strong Empirical Support

- **Low-Score Goal Dependency**: Empirical analysis across 30+ years of European league data confirms that independent Poisson models underestimate 0-0 and 1-1 draws by 10\% \text{ to } 15\%. The Dixon-Coles \tau correction and bivariate Poisson formulations resolve this error.
- **Closing Odds Benchmark**: Extensive empirical financial studies confirm that bookmaker closing odds display higher predictive accuracy (lower log loss) than opening odds or static statistical models, validating CLV as a benchmark for quantitative edge.
- **Shin's De-vigging Superiority**: Empirical tests verify that Shin's method removes favourite-longshot bias more effectively than proportional normalization, yielding fair probabilities that align closely with closing prices.

3. Mixed or Context-Dependent Evidence

- **Machine Learning vs Structural Goal Models**: Claims that complex machine learning models (e.g., XGBoost, Deep Neural Networks) systematically beat structural statistical models (Dixon-Coles, Bayesian Hierarchical Bivariate Poisson) are mixed. While ML models capture non-linear interactions in large feature sets, they frequently produce uncalibrated probabilities on tail scorelines and require post-hoc calibration to match structural Poisson reliability.
- **Granular Tracking Data Value**: Granular spatial tracking data (25\text{ Hz} player coordinates) improves local shot xG estimates, but when aggregated to full-match result forecasting, the predictive gain over event-based xG and team strength metrics is marginal relative to data acquisition costs.

4. Unverified Claims and Speculation

- **Guaranteed Model Edges**: Claims found in commercial betting literature that simple uncalibrated Poisson models guarantee positive expected returns are false. Bookmaker overround (M \approx 3\% - 7\%) easily absorbs naive model variances.
- **Full Kelly Staking Profitability**: Claims that bettors should execute full Kelly staking are dangerous and speculative; real-world parameter estimation error causes full Kelly staking to induce catastrophic drawdowns (>70\%) with high probability.

Deliverable F: Proposed Mathematical Framework

The FootyQuant v2 core architecture is organized as a 9-stage sequential probabilistic pipeline:

1. **Stage 1: Raw Data & Feature Pipeline**: Ingests historic and live scorelines, event coordinate logs, and bookmaker decimal odds streams.
2. **Stage 2: Latent Ability & Metric Engine**: Extracts Expected Goals (xG) shot quality and Expected Threat (xT) spatial ball progression metrics, updating dynamic Elo and state-space team strength ratings.
3. **Stage 3: Structural Score Distribution Model**: Evaluates goal distributions via Dixon-Coles or Bayesian Bivariate Poisson formulations using [span_283](start_span)[span_283](end_span)[span_284](start_span)[span_284](end_span)[span_285](start_span)[span_285](end_span)footBayes MCMC sampling.
4. **Stage 4: Probability Matrix & Market Mapping**: Performs exact matrix summation over scoreline distributions to calculate raw probabilities for 1X2, Over/Under, Both Teams to Score, and Asian Handicap markets.
5. **Stage 5: Calibration & Uncertainty Quantification**: Applies Beta Calibration trained on historical out-of-sample splits and quantifies parameter epistemic variance across MCMC draws.
6. **Stage 6: Market De-vigging & Consensus Alignment**: Processes multi-bookmaker decimal odds using Shin's insider trading model to extract fair market consensus probabilities.
7. **Stage 7: Robust Expected Value & Abstention Decision Filter**: Evaluates expected value EV = P_{\text{cal}} \cdot O - 1 under a dual-threshold rule, rejecting bets where epistemic variance exceeds safety limits.
8. **Stage 8: Portfolio Joint Optimization & Staking Engine**: Solves the multi-asset Kelly matrix optimization problem (\mathbf{f}^* = \boldsymbol{\Sigma}^{-1} \mathbf{e}), scaling stakes by fraction c = 0.25 and enforcing exposure limits.
9. **Stage 9: Validation, CLV Tracking & Execution Monitor**: Logs real-time market execution, tracks Closing Line Value realization, and monitors out-of-sample log-loss calibration drift.

Component Interactions and Error Propagation

1. **Inputs**: Raw historic and live match event data, shot logs, team lineups, and closing odds feeds enter Stage 1.
2. **Latent Metrics**: Stage 2 extracts xG and xT spatial progression matrices, updating Bayesian dynamic team attack (\alpha_i) and defense (\beta_i) parameters.

3. **Score Matrix Generation**: Stage 3 inputs team parameters into a Dixon-Coles or Bivariate Poisson structural model, outputting the discrete scoreline probability mass matrix P(X=x, Y=y).
4. **Market Transformation**: Stage 4 aggregates joint matrix cells to produce raw model market probabilities P_{\text{raw}}(\text{Market}_k).
5. **Calibration & Uncertainty**: Stage 5 applies Beta Calibration (trained on historical out-of-sample splits). It calculates parameter epistemic variance \sigma[span_76](start_span)[span_76](end_span)_{\text{epi[span_35](start_span)[span_35](end_span)[span_52](start_span)[span_52](end_span)stemic}}^2 across Bayesian MCMC posterior draws.
6. **Market Comparison**: Stage 6 ingests multi-bookmaker decimal odds, applies Shin's method to extract fair market consensus probabilities p_{\text{market}, k}, and evaluates Opening vs Closing Line Value.
7. **Decision Filtering**: Stage 7 computes robust expected value EV = P_{\text{cal}} \cdot O - 1. If EV < EV_{\min} or epistemic variance \sigma_{\text{epistemic}}^2 > \sigma_{\max}^2, the selection is routed to **Abstain (No Bet)**.
8. **Capital Allocation**: Stage 8 passes all active positive EV opportunities to the Multi-Asset Kelly matrix optimizer (\mathbf{f}^* = \boldsymbol{\Sigma}^{-1} \mathbf{e}), scaling stakes by fraction c = 0.25 and enforcing maximum match exposure constraints.
9. **Execution Monitoring**: Stage 9 logs executed bets, tracks CLV realization, and monitors log-loss calibration drift.

Deliverable G: Experimental Validation Protocol

To evaluate FootyQuant v2 without dataset shift, look-ahead bias, or data snooping, all modeling components must adhere to a strict chronological walk-forward validation protocol. The walk-forward evaluation protocol uses a rolling temporal structure:

1. **Training Window**: Expanding or rolling 3-season temporal block (e.g., T-3 to T-1, containing \approx 1140 matches per league) used to train Bayesian dynamic team strength parameters (\alpha_i, \beta_i) and structural goal models.
2. **Validation Block**: Immediate preceding 500 chronological matches used strictly to fit post-hoc Beta Calibration curves and optimize decision thresholds (EV_{\min}, \sigma_{\max}^2).
3. **Evaluation Step**: Out-of-sample evaluation performed exclusively on matchday t. Predictions are generated, odds recorded, and execution logged.
4. **Temporal Roll**: Advance time window by 1 matchday (t \to t+1); re-estimate team parameters sequentially without backward information flow.

1. Temporal Data Splitting

- **No Random Cross-Validation**: Standard K-fold random cross-validation is strictly prohibited due to temporal autocorrelation and leakage of dynamic team strength information.
- **Rolling Origin Design**:
    - **Initial Training Set**: Seasons T-3 to T-1 (e.g., 3 complete historical seasons, \approx 1140 matches per league).
    - **Calibration Set**: The immediate preceding 500 chronological matches prior to

matchday t.
    - **Evaluation Set**: Matchday t.
    - **Step Function**: Advance time window by 1 matchday; re-estimate dynamic team parameters sequentially.

2. Strict Anti-Leakage Rules

1. **Timestamp Enforcement**: All model feature values must be computed using data recorded strictly prior to market execution time t_{\text{exec}}.
2. **Lineup Availability Constraints**: Unofficial predicted lineups cannot use post-publication actual squad parameters. If official team sheets are published at t - 60 \text{ mins}, execution evaluating lineup shock must occur strictly within t \in [t-60, t-15].
3. **Closing Odds Separation**: Closing odds O_{\text{closing}} are used strictly as an evaluation benchmark in Stage 9 and must never enter Stage 1–5 feature engineering.

3. Primary Evaluation Metrics

Validation requires evaluating probabilistic performance and market execution independently:

1. **Log Loss (Out-of-Sample)**: Primary probabilistic metric. Measures distribution distance: \text{Log Loss} = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K y_{ik} \ln \hat{p}_{ik}
2. **Expected Calibration Error (ECE)**: Measures calibration reliability across M=10 equal-width probability bins: \text{ECE} = \sum_{j=1}^M \frac{|B_j|}{N} \left| \text{acc}(B_j) - \text{conf}(B_j) \right|
3. **CLV Beat Rate**: Proportion of wagers where executed odds exceed final fair closing odds: \text{CLV Rate} = \frac{1}{N_{\text{bets}}} \sum_{i=1}^{N_{\text{bets}}} I\left(O_{\text{exec}, i} > O_{\text{fair, closing}, i} \right)
4. **Realized Yield (Return on Investment - ROI)**: \text{ROI} = \frac{\sum_{i=1}^{N_{\text{bets}}} \text{Payout}_i - \sum_{i=1}^{N_{\text{bets}}} \text{Stake}_i}{\sum_{i=1}^{N_{\text{bets}}} \text{Stake}_i}
5. **Max Drawdown (MDD)**: MDD = \max_{t \in [0, T]} \left(\frac{\max_{\tau \in [0, t]} W_\tau - W_t}{\max_{\tau \in [0, t]} W_\tau} \right)

Deliverable H: Prioritized Learning Roadmap

To master the quantitative disciplines underlying FootyQuant v2, researchers must progress through a structured 4-phase curriculum spanning probability theory, goal modeling, market microstructure, and capital allocation, as outlined below:

Phase 1: Measure Theory, Probability, and Frequentist Foundations

- **Prerequisites**: Multivariable calculus, linear algebra.
- **Core Topics**: Probability spaces (\Omega, \mathcal{F}, P), Kolmogorov axioms, discrete and continuous random variables, joint distributions, expectation, variance, covariance, Maximum Likelihood Estimation (MLE), L1/L2 regularization, Shannon entropy, cross-entropy, proper scoring rules.
- **Primary Learning Source**: DeGroot & Schervish, *Probability and Statistics* (4th Edition).

Phase 2: Football Outcome Modelling and Spatial Analytics

- **Prerequisites**: Phase 1 mastery, Poisson processes, generalized linear models (GLMs).
- **Core Topics**: Independent Poisson scoreline models (Maher), Dixon-Coles low-score dependency corrections (\tau) and time decay (\phi), Bivariate Poisson models (Karlis & Ntzoufras), Skellam goal-difference formulations, Bayesian hierarchical modeling in Stan (footBayes), Expected Goals (xG) logistic modeling, Expected Threat (xT) spatial grid Bellman iteration.
- **Primary Learning Source**: Egidi, Karlis, & Ntzoufras, *Predictive Modelling for Football Analytics* (CRC Press, 2026).

Phase 3: Probability Calibration and Betting Market Microstructure

- **Prerequisites**: Phase 2 mastery, Bayesian inference, microeconomics.
- **Core Topics**: Brier score decomposition (Reliability, Resolution, Uncertainty), post-hoc calibration (Platt scaling, Isotonic regression, Beta calibration), bookmaker overround structures, margin removal algorithms, Shin's insider trading model, Efficient Market Hypothesis in sports, Closing Line Value (CLV) evaluation, favourite-longshot bias quantification.
- **Primary Learning Source**: Shin (1993), "Optimal Betting Odds against Insider Traders", *The Economic Journal*.

Phase 4: Capital Allocation, Portfolio Optimization, and Stochastic Control

- **Prerequisites**: Phase 3 mastery, real analysis, constrained optimization.
- **Core Topics**: Von Neumann-Morgenstern utility theory, logarithmic wealth growth optimization, formal derivation of the Kelly criterion, Fractional Kelly under parameter uncertainty, continuous Risk of Ruin approximations, multi-asset portfolio Kelly matrix optimization (\mathbf{f}^* = \boldsymbol{\Sigma}^{-1} \mathbf{e}), intra-match market return covariance calculation, dual-threshold robust decision policies.
- **Primary Learning Source**: MacLean, Thorp, & Ziemba, *The Kelly Capital Growth Investment Criterion* (World Scientific).

Deliverable I: Open Research Questions

To advance FootyQuant v2 beyond existing academic baselines, the research team must address four open scientific questions:

1. **Non-Stationary In-Play Hazard Intensities Under Structural Disturbances**: Current in-play Markov scoring models assume smooth intensity shifts following red cards or goals. How can structural jump-diffusion processes or hawkes point processes be parameterised to model immediate post-event tactical volatility (e.g., scoring intensity within 5 \text{ minutes} of a goal or red card)?
2. **Optimal Parameterization of Epistemic Variance Penalties in Multi-Asset Kelly Staking**: While fractional Kelly (c=0.25) reduces drawdown risk, it treats parameter uncertainty as a static multiplier. What is the closed-form solution for the portfolio Kelly vector \mathbf{f}^* when individual excess returns e_i are random variables with known posterior variance distributions \sigma_{\text{epistemic}, i}^2?
3. **Cross-League Hierarchical Parameter Transfer for Promoted and Relegated Teams**:

Teams promoted from lower divisions (e.g., EFL Championship to English Premier League) lack historical match data against top-flight opponents. How can Bayesian structural transfer learning quantify league quality step-changes to minimize parameter estimation variance during the first 10 \text{ matchdays} of a season?

4. **Distributionally Robust Market Margin Extraction Under Non-Uniform Insider Trading**: Shin's de-vigging model assumes a single uniform insider fraction z across all market selections. How can Shin's framework be extended to multi-runner Asian Handicap or Over/Under lines where insider information is concentrated asymmetric on specific total goal thresholds?

Deliverable J: Research-to-Implementation Specification

This specification defines the formal mathematical interfaces, equations, data schemas, and transformations required to transition FootyQuant v2 into technical development.

1. Mathematical Data Contracts & Schemas

Input Interface: Match Fixture Data (FixtureInput)

- fixture_id: Unique identifier (String)
- home_team_id: Unique identifier (String)
- away_team_id: Unique identifier (String)
- timestamp: Epoch seconds t_{\text{exec}} (Integer)
- league_id: Unique identifier (String)
- historical_scorelines: Vector of historical tuples (x_k, y_k, t_k)

Intermediate Data Contract: Structural Parameters (ModelParameters)

- alpha_home: Home attacking strength \alpha_i \in \mathbb{R}
- beta_home: Home defensive weakness \beta_i \in \mathbb{R}
- alpha_away: Away attacking strength \alpha_j \in \mathbb{R}
- beta_away: Away defensive weakness \beta_j \in \mathbb{R}
- gamma: Home advantage parameter \gamma \in \mathbb{R}^+
- rho: Dixon-Coles dependency parameter \rho \in (-0.25, 0.25)
- xi: Temporal decay rate parameter \xi \in \mathbb{R}^+

Output Interface: Score Mass Matrix (ScoreMatrixOutput)

- matrix: 16 \times 16 floating-point array P(X=x, Y=y), where \sum_{x=0}^{15} \sum_{y=0}^{15} P(x, y) = 1.0
- epistemic_variance_matrix: 16 \times 16 array containing posterior variances \text{Var}_{\text{posterior}}[span_80](start_span)[span_80](end_span)(P(x, y)) across MCMC draws

2. Core Operational Equations & Transformations

Step 1: Compute Expected Goal Intensities

\lambda_{ij} = \exp\left(\mu + \gamma + \alpha_i + \beta_j \right) \mu_{ij} = \exp\left(\mu + \alpha_j + \beta_i \right)

Step 2: Populate Dixon-Coles Joint Mass Matrix

For x \in \{0, \dots, 15\} and y \in \{0, \dots, 15\}:
P(X=x, Y=y) = \tau_{\rho, \lambda_{ij}, \mu_{ij}}(x, y) \left(\frac{\lambda_{ij}^x e^{-\lambda_{ij}}}{x!} \right) \left(\frac{\mu_{ij}^y e^{-\mu_{ij}}}{y!} \right)
Where \tau correction applies to (x,y) \in \{0,1\}^2 per Deliverable B, Section 2.

Step 3: Map Score Matrix to Raw Market Probabilities

P_{\text{raw}}(\text{Home}) = \sum_{x > y} P(x, y) P_{\text{raw}}(\text{Draw}) = \sum_{x = y} P(x, y) \\ P_{\text{raw}}(\text{Away}) = \sum_{x < y} P(x, y)
P_{\text{r[span_130](start_span)[span_130](end_span)aw}}(\text{Over 2.5}) = \sum_{x+y \ge 3} P(x, y)

Step 4: Probability Calibration Transformation

For each market probability r = P_{\text{raw}}:
P_{\text{cal}} = \frac{1}{1 + \exp\left(-\left(a \ln(r) + b \ln(1 - r) + c \right) \right)}
Where parameters (a, b, c) are loaded from the stored Beta Calibration artifact.

Step 5: Extract Fair Odds via Shin's De-vigging Method

Given raw market odds (O_1, O_2, O_3), compute implied probabilities \pi_k = \frac{1}{O_k} and solve for insider fraction z:
\sum_{k=1}^3 \left(\frac{\sqrt{z^2 + 4(1-z) \frac{\pi_k}{\sum \pi_j}} - z}{2(1-z)} \right) = 1
Recover fair market probability p_{\text{fair}, k}:

p_{\text{fair}, k} = \frac{\sqrt{z^2 + 4(1-z) \frac{\pi_k}{\sum \pi_j}} - z}{2(1-z)}

Step 6: Expected Value & Abstention Evaluation

EV_k = P_{\text{cal}, k} \cdot O_{\text{exec}, k} - 1
Execution Decision Policy:

\text{Decision}_k = \begin{cases} \text{PASS} & \text{if } EV_k \le 0.03 \\ \t[span_132](start_span)[span_132](end_span)ext{PASS} & \text{if } \text{Var}_{\text{epistemic}}(P_k) > 0.0025 \\ \text{BET} & \text{if } EV_k > 0.03 \quad \text{AND} \quad \text{Var}_{\text{epistemic}}(P_k) \le 0.0025 \end{cases}

Step 7: Portfolio Multi-Asset Stake Sizing

For set of M approved wagers with return vector \mathbf{e} and covariance matrix \boldsymbol{\Sigma}:
\mathbf{f}^* = 0.25 \cdot \boldsymbol{\Sigma}^{-1} \mathbf{e}
Enforce individual stake cap f_i \le 0.05 (5\% max bankroll per wager) and aggregate match exposure cap \sum_{i \in \text{Match}} f_i \le 0.10 (10\% max bankroll per fixture).

3. Execution Interface Contract (StakeAllocationOutput)

- fixture_id: Unique identifier (String)
- market_type: Market identifier (e.g., 1X2_HOME, TOTAL_OVER_2.5)
- calibrated_probability: P_{\text{cal}} \in [0, 1]
- fair_closing_odds: O_{\text{fair}} = \frac{1}{p_{\text{fair}}}
- expected_value: Calculated EV \in \mathbb{R}
- epistemic_variance: Measured parameter variance \sigma_{\text{epistemic}}^2
- recommended_stake_fraction: Final portfolio fraction f_i^* \in [0, 0.05]
- recommended_stake_currency: Absolute stake amount in bankroll currency (e.g., \text{₹}12.50 for W_0[span_81](start_span)[span_81](end_span) = \text{₹}500)

Works cited

1. Maher 1982 Poisson Model: Paper, Formula & Results | ImpliedScore, https://impliedscore.com/maher-football-poisson-model/ 2. Modelling association football scores and inefficiencies in the, https://research.lancaster-university.uk/en/publications/modelling-association-football-scores-and-inefficiencies-in-the-f/ 3. Skellam distribution - Wikipedia, https://en.wikipedia.org/wiki/Skellam_distribution 4. Post-hoc calibration without distributional assumptions - Chirag Gupta, https://aigen.github.io/files/proposal.pdf 5. (PDF) Aleatoric and epistemic uncertainty in machine learning, https://www.researchgate.net/publication/349905827_Aleatoric_and_epistemic_uncertainty_in_machine_learning_an_introduction_to_concepts_and_methods 6. Comment nous prédisons la Coupe du Monde 2026 ... - OddsFlow.ai, https://www.oddsflow.ai/fr/worldcup/methodology 7. A Study of the Kelly Criterion: from Betting to Financial Markets, https://diposit.ub.edu/bitstreams/b11028ae-ab78-4bf3-a22b-1cbb3821ad3f/download 8. Kelly criterion - Wikipedia, https://en.wikipedia.org/wiki/Kelly_criterion 9. Kelly Criterion Explained: Size Every Trade for Growth - MoneyFlock, https://www.moneyflock.com/contents/articles/kelly-criterion-position-sizing 10. Uncertainty in the Binomial Game - OU Math, https://math.ou.edu/~pbretz/Uncertainty_in_the_Binomial_Game.pdf 11. (PDF) footBayes: fitting football models and visualizing predictions, https://www.researchgate.net/publication/362155489_footBayes_fitting_football_models_and_visualizing_predictions_with_the_footBayes_package 12. Dixon-Coles Model Explained: Rho, Tau & Football Scores, https://impliedscore.com/dixon-coles-model/ 13. Alternative ranking measures to predict international football results, https://arts.units.it/retrieve/7a13757e-7dc2-4bec-bfd3-faea2f0952ea/Alternative_ranking_measures_to_predict_internatio-Post_print.pdf 14. Skellam Distribution (Poisson Difference Distribution) - Statistics, https://www.statisticshowto.com/skellam-distribution/ 15. (PDF) Bayesian modelling of football outcomes: using the Skellam's, https://www.researchgate.net/publication/228621612_Bayesian_modelling_of_football_outcomes_using_the_Skellam's_distribution_for_the_goal_difference 16. Bayesian weighted discrete-time dynamic models for association, https://academic.oup.com/jrsssc/advance-article/doi/10.1093/jrsssc/qlag032/8704597 17. Predictive Modelling For Football Analytics | PDF - Scribd, https://www.scribd.com/document/1067241883/Predictive-Modelling-for-Football-Analytics 18.

Expected Threat (xT) Soccer Calculator - MetricGate, https://metricgate.com/docs/expected-threat-xt-soccer/ 19. What is Expected Threat (xT)? Possession Value models explained, https://www.hudl.com/blog/possession-value-models-explained 20. Expected Threat (xT) the best offensive metric so far, https://www.footballytics.ch/post/expected-threat-xt-the-best-offensive-metric-so-far 21. Expected Threat (xT) — DataBallPy Documentation - Read the Docs, https://databallpy.readthedocs.io/en/main/features/xt_models.html 22. Understanding Expected Threat (xT) in Football Betting: How PICA, https://picatip.com/blog/understanding-expected-threat-xt-football-betting-pica-ai-advanced-metric-sportybet 23. Modelling Association Football Scores and Inefficiencies in the, https://academic.oup.com/jrsssc/article-abstract/46/2/265/6990546 24. Football Betting - Statistics, http://www.stats.org.uk/football-betting/ 25. Analysis of sports data using bivariate Poisson models | Request PDF, https://www.researchgate.net/publication/227719079_Analysis_of_sports_data_using_bivariate_Poisson_models 26. Comparison of Consensus and Roberts Assessments | Download, https://www.researchgate.net/figure/Comparison-of-Consensus-and-Roberts-Assessments_fig10_220210248 27. Forecasting football results and the efficiency of fixed-odds betting, https://ptacts.uspto.gov/ptacts/public-informations/petitions/1550668/download-documents?artifactId=84oQ8ifIwsbybua5CtMhDKRUtPt1XFnlm_yNlSycBplx0PLsJy8Pmzw